

Nexivarin 10 mg Tablets — eCTD Submission  
Module 4 — Nonclinical Study Reports

|                 |                 |                 |                              |
|-----------------|-----------------|-----------------|------------------------------|
| Submission Type | NDA — 505(b)(1) | Sequence Number | 0000                         |
| Submission Date | 2021-03-15      | Section         | <a href="#">4.2.1.3</a>      |
| Prepared by     | PharmaCorp Inc. | Confidentiality | Proprietary and Confidential |

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| Section                 | Title                                     | Description                                                |
|-------------------------|-------------------------------------------|------------------------------------------------------------|
| <a href="#">4.2.1.3</a> | Safety Pharmacology                       | ICH S7A/S7B core battery and follow-up studies             |
| Study<br>NXV-SP-001     | hERG Channel Inhibition (In Vitro)        | Patch-clamp; IC50 >30 µM — no proarrhythmic liability      |
| Study<br>NXV-SP-002     | Cardiovascular Safety (Telemetered Dog)   | Hemodynamics and ECG monitoring; vasodilatory effects only |
| Study<br>NXV-SP-003     | CNS Safety Pharmacology (Irwin)           | No CNS effects up to 100 mg/kg                             |
| Study<br>NXV-SP-004     | Respiratory Safety (Plethysmography, Rat) | No adverse respiratory effects                             |

PharmaCorp Inc.  
NDA 123456

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Section 4.2.1.3  
Safety Pharmacology

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### 4.2.1.3 Safety Pharmacology — Nexivarin Hydrochloride

Safety pharmacology studies were conducted in accordance with ICH S7A (Core Battery) and ICH S7B (hERG, QT) guidelines.

#### Cardiovascular Safety Pharmacology (ICH S7B)

| Study                                              | System                     | Finding                                                            | Conclusion                    |
|----------------------------------------------------|----------------------------|--------------------------------------------------------------------|-------------------------------|
| hERG channel inhibition (in vitro, Xenopus oocyte) | hERG (IKr) patch clamp     | IC50 >10,000 nM (>4300-fold safety margin vs. Cmax 2.3 nM)         | Negligible hERG inhibition    |
| Action potential duration (rabbit Purkinje fiber)  | Ventricular APD90          | No significant change at 100× Cmax                                 | No proarrhythmic risk         |
| Cardiac hemodynamics (anesthetized dog, telemetry) | HR, BP, ECG (PQ, QRS, QTc) | No significant QTc change at 3, 10, 30 mg/kg IV; mild BP reduction | No cardiac conduction effects |
| In vivo telemetry (conscious dog)                  | Holter ECG, BP, HR         | No arrhythmias; BP reduction dose-dependent at 3 and 10 mg/kg      | Consistent with mechanism     |

#### CNS Safety Pharmacology (ICH S7A Core Battery)

| Study                        | Species | Doses                 | Key Findings                                                                       |
|------------------------------|---------|-----------------------|------------------------------------------------------------------------------------|
| Modified Irwin Observation   | Mouse   | 30, 100, 300 mg/kg PO | No adverse CNS signs at any dose; mild sedation at 300 mg/kg (10× MRHD equivalent) |
| Locomotor Activity           | Mouse   | 30, 100, 300 mg/kg PO | No statistically significant change in locomotor activity up to 300 mg/kg          |
| Barbiturate Sleep Time       | Mouse   | 100, 300 mg/kg PO     | No potentiation of barbiturate-induced sleep                                       |
| Pentylenetetrazol Convulsion | Mouse   | 100, 300 mg/kg PO     | No anti- or pro-convulsant activity                                                |

#### Respiratory Safety Pharmacology (ICH S7A Core Battery)

| Study                      | Species | Doses                 | Key Findings                                                               |
|----------------------------|---------|-----------------------|----------------------------------------------------------------------------|
| Whole-body plethysmography | Rat     | 30, 100, 300 mg/kg PO | No significant effects on respiratory rate, tidal volume, or minute volume |
| Blood gas analysis         | Rat     | 100 mg/kg IV          | No change in PaO2, PaCO2, pH vs. vehicle at 0.5, 1, 2 h                    |

**Overall Safety Pharmacology Conclusion:** Nexivarin hydrochloride does not present any safety signals in CNS, cardiovascular, or respiratory systems at exposures substantially in excess of the therapeutic range. The cardiovascular findings (mild BP reduction) are consistent with the intended pharmacological mechanism and are not unexpected.